

Transfer Learning for Binary Facial Emotion Classification: A Comparative Evaluation of ResNet18 and MobileNetV2

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Abstract—Transfer learning has become a widely adopted approach for image classification tasks where labeled data are limited. This study presents a comparative evaluation of ResNet18 and MobileNetV2 for binary facial emotion classification using image frames extracted from the RAVDESS dataset. The original emotion categories were reformulated into a binary problem consisting of “happy” and “not happy” classes. A balanced dataset of 2,332 images was divided into training/validation and independent test subsets, and model performance was assessed using stratified 5-fold cross-validation. Both architectures were initialized with pretrained ImageNet weights and fine-tuned using transfer learning techniques. Performance was evaluated using accuracy, precision, recall, F1-score, and validation loss.

Experimental results show that both models achieved strong classification performance, with average validation accuracies exceeding 94% and test accuracies exceeding 96%. MobileNetV2 achieved the highest overall performance, obtaining an average validation accuracy of 95.23% and a test accuracy of 96.36%, while ResNet18 achieved an average validation accuracy of 94.96% and a test accuracy of 96.15%. MobileNetV2 also achieved the highest F1-score (0.9642) and the lowest test loss, indicating slightly better generalization performance. These results demonstrate that lightweight architectures can achieve performance comparable to deeper residual networks while offering advantages for deployment in resource-constrained environments. The findings further support the effectiveness of transfer learning and modern convolutional neural network architectures for facial emotion classification tasks.

Keywords: Deep learning, binary image classification, transfer learning, ResNet18, MobileNetV2, K-fold cross-validation, convolutional neural networks, image classification, performance evaluation, model comparison, computer vision, deep neural networks, feature learning.

I. INTRODUCTION

The rapid adoption of computer vision technologies in real-world environments has increased the demand for deep learning models capable of delivering high classification accuracy while operating under limited computational resources [1], [2]. Applications including intelligent surveillance, human-computer interaction, healthcare monitoring, mobile devices, and embedded edge systems often require real-time image analysis without access to high-performance computing infrastructure [2]. In these scenarios, computational efficiency

becomes as important as predictive performance, motivating the need to evaluate architectures that balance classification accuracy, model complexity, and deployment feasibility.

Recent research further emphasizes the growing importance of efficient deep learning models for edge deployment and real-time inference, where latency and energy constraints significantly influence model selection [3]. At the same time, modern studies highlight that architectural optimization and transfer learning remain key enablers for deploying high-performing vision systems in resource-constrained environments [4].

Facial emotion recognition represents a practical computer vision task in which reliable classification performance is essential for understanding human affective states and supporting intelligent decision-making systems [5]. Recent advances in transfer learning have enabled convolutional neural networks (CNNs) pretrained on large-scale image datasets to achieve strong performance even when task-specific datasets are relatively small [6], [7]. However, the trade-off between model complexity and predictive accuracy remains an important consideration, particularly for edge and embedded vision applications where memory usage, power consumption, and inference latency are constrained [2].

More recent surveys in facial expression recognition confirm that deep CNN-based approaches continue to dominate state-of-the-art performance, while lightweight architectures are increasingly preferred for practical deployment scenarios [8]. This reinforces the need to systematically compare high-capacity models with efficient alternatives under identical experimental conditions.

This study investigates whether lightweight neural network architectures can achieve performance comparable to deeper and more computationally intensive models in a binary facial emotion classification task. Specifically, ResNet18 [9] and MobileNetV2 [10] were selected because they represent two widely adopted design philosophies in modern computer vision: deep residual learning and efficient mobile-oriented network design. Both models were initialized with pretrained ImageNet weights [7] and fine-tuned using transfer learning

on image frames extracted from the RAVDESS dataset [11]. The original emotion labels were reformulated into a binary classification problem consisting of “happy” and “not happy” categories.

Model performance was evaluated using stratified 5-fold cross-validation together with an independent holdout test set. Performance metrics included classification accuracy, precision, recall, F1-score, validation loss, and confusion matrices to provide a comprehensive assessment of each architecture. By comparing both models under identical experimental conditions, this work provides a reproducible evaluation of their effectiveness for facial emotion classification while examining the practical trade-offs between predictive performance and computational efficiency. The findings contribute to ongoing research in resource-efficient deep learning and support the development of computer vision systems suitable for deployment in real-world edge computing environments.

II. RELATED WORK

A. Facial Emotion Recognition

Facial emotion recognition (FER) is a well-studied problem in computer vision with applications in human–computer interaction, surveillance, and affective computing [5]. Traditional FER approaches relied on handcrafted feature extraction methods such as Local Binary Patterns (LBP) and Histogram of Oriented Gradients (HOG), combined with classical machine learning classifiers. However, these methods often exhibit limited robustness under variations in pose, illumination, and occlusion.

The introduction of deep learning significantly improved FER performance by enabling automatic feature learning through convolutional neural networks (CNNs). These models learn hierarchical representations of facial expressions directly from raw images, improving generalization across different conditions.

B. Transfer Learning in Deep Neural Networks

Transfer learning has become a standard technique in deep learning, particularly for image classification tasks with limited labeled data [6]. By leveraging models pretrained on large-scale datasets such as ImageNet [7], neural networks can reuse learned feature representations and adapt them to new tasks through fine-tuning.

This approach reduces training time, improves convergence, and mitigates overfitting, especially in domains where labeled datasets are limited, such as medical imaging and facial emotion recognition.

C. Deep Learning Architectures for Vision Tasks

Convolutional neural networks have evolved significantly in both accuracy and efficiency. ResNet18 introduces residual connections that address the vanishing gradient problem and enable deeper network training [9]. It is widely used due to its strong performance and relatively moderate computational cost.

MobileNetV2 is designed for mobile and embedded applications, using depthwise separable convolutions and inverted residual blocks to reduce computational complexity while maintaining competitive accuracy [10]. This makes it particularly suitable for real-time and resource-constrained environments.

D. Efficient Deep Learning and Computational Constraints

Modern deep learning systems must balance accuracy with computational efficiency. Efficient neural network design is critical for deployment on edge devices, where memory, power consumption, and latency are constrained [2]. As a result, lightweight architectures such as MobileNetV2 have become increasingly relevant for real-world applications.

E. Dataset and Benchmarking Context

The RAVDESS dataset provides a standardized benchmark for emotion-related machine learning tasks, offering controlled facial and vocal expressions for evaluation [11]. It has been widely used in research on facial and audio emotion recognition due to its structured design and high-quality annotations.

III. METHODOLOGY

This study employed a supervised deep learning framework to compare the performance of two convolutional neural network (CNN) architectures, ResNet18 and MobileNetV2, for binary facial emotion classification using transfer learning. The complete experimental pipeline consisted of dataset preparation, image preprocessing, transfer learning, stratified five-fold cross-validation, and independent holdout testing. Figure 1 illustrates the overall workflow.

A. Dataset Preparation

Experiments were conducted using image frames extracted from the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) [11]. Rather than using the original multimodal recordings, only facial image frames were considered for visual emotion recognition.

The original multi-class emotion labels were converted into a binary classification problem. Images corresponding to the “happy” emotion were assigned to the positive class, while all remaining emotions were grouped into the “not happy” class. To reduce class bias, the majority class was randomly downsampled to match the minority class, resulting in a balanced dataset containing 2,332 images.

The balanced dataset was divided using an 80/20 stratified split, producing 1,865 images for model development and 467 images for independent testing. The independent test set remained completely unseen during training and validation.

B. Image Preprocessing and Data Augmentation

Image preprocessing was implemented using the PyTorch and Torchvision libraries [12]. All images were resized to the input resolution required by the pretrained CNN architectures and normalized using the ImageNet mean and standard deviation [7].

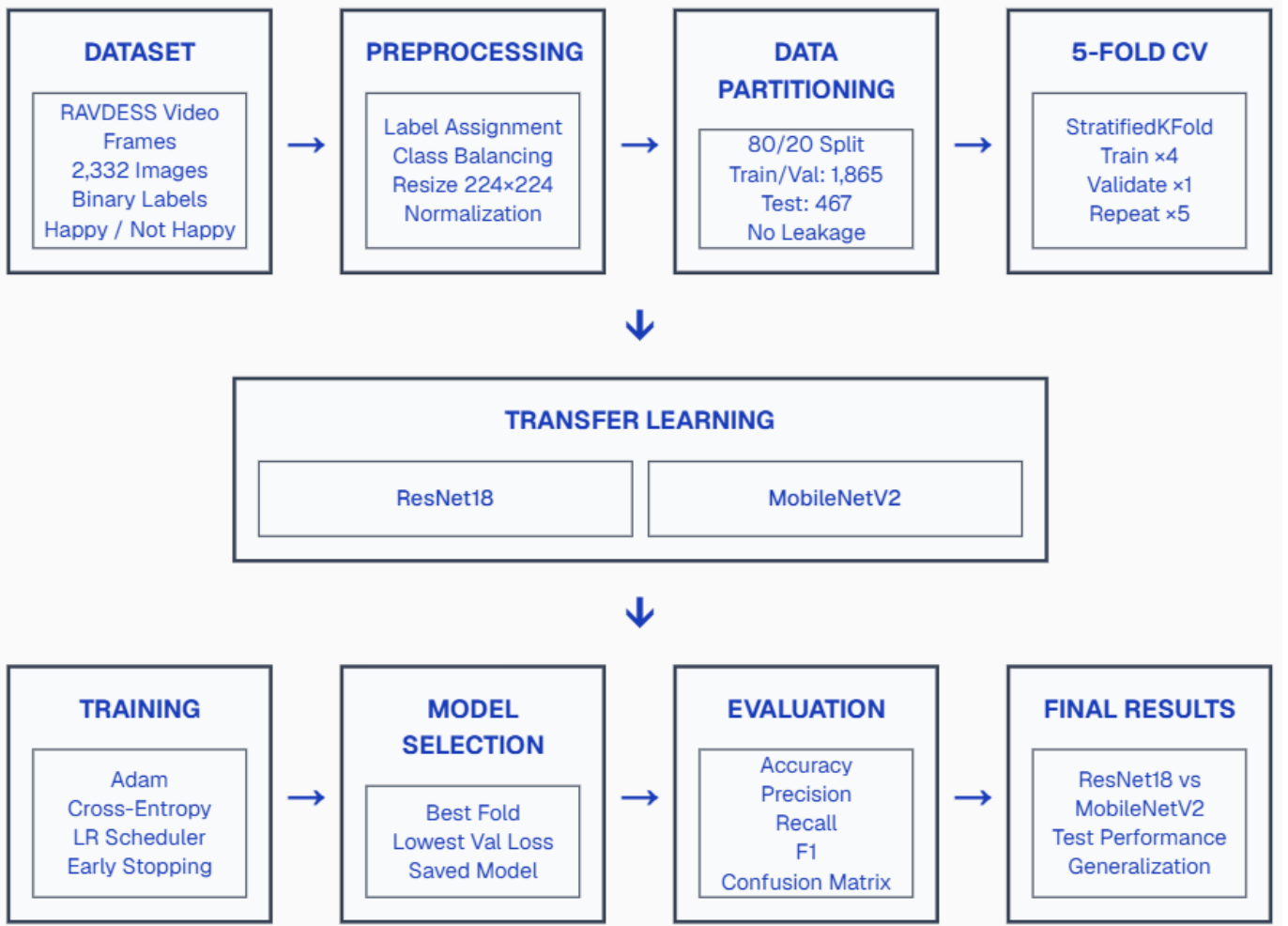


Fig. 1. Overall workflow of the proposed transfer learning framework, including data preprocessing, dataset partitioning, transfer learning with ResNet18 and MobileNetV2, stratified 5-fold cross-validation, and final model evaluation.

To improve model generalization and reduce overfitting, data augmentation was applied exclusively to the training subset. Augmentation operations included random horizontal flipping, small random rotations, and color jittering. Validation and test images received only deterministic resizing and normalization to ensure unbiased performance evaluation.

C. Transfer Learning Models

Two pretrained CNN architectures were evaluated: ResNet18 [9] and MobileNetV2 [10]. Both networks were initialized using ImageNet pretrained weights and fine-tuned for binary classification using transfer learning [6].

For each architecture, the original classification layer was replaced with a fully connected layer containing two output neurons corresponding to the “happy” and “not happy” classes. All remaining network parameters were optimized during fine-tuning.

D. Training Procedure

Model development was performed in Python using the PyTorch deep learning framework [12]. Training was conducted

on the training/validation subset using stratified five-fold cross-validation. Stratification ensured that each fold preserved the original class distribution, providing reliable estimation of model generalization performance [13].

For each fold, four partitions were used for training and one partition was reserved for validation. Models were trained for a maximum of 15 epochs using the Adam optimization algorithm [14] with an initial learning rate of 1×10^{-4} and a batch size of 32. Cross-entropy loss was used as the optimization objective.

A ReduceLROnPlateau learning rate scheduler automatically reduced the learning rate when validation loss stopped improving. Early stopping with a patience of three epochs was employed to prevent overfitting. The model achieving the lowest validation loss within each fold was retained.

After completion of all five folds, the model with the best overall validation loss was selected as the final model for evaluation on the independent holdout test set.

E. Performance Evaluation

Model performance was evaluated using both cross-validation and an independent test set. During each validation fold, classification performance was measured using accuracy, precision, recall, and F1-score. Confusion matrices and detailed classification reports were also generated to analyze prediction behavior.

Following completion of cross-validation, the selected best-performing model was evaluated on the independent holdout test set containing 467 previously unseen images. The same evaluation metrics were computed to estimate real-world generalization performance.

All evaluation metrics, including accuracy, precision, recall, F1-score, confusion matrices, and classification reports, were computed using the Scikit-learn machine learning library [15]. Final model performance was reported as the average validation metrics across the five folds together with the independent test results, providing a comprehensive assessment of classification accuracy, robustness, and generalization capability.

IV. EXPERIMENTAL SETUP

The experiments were implemented in the Python programming language using the PyTorch deep learning framework [12]. Model development and training were performed within the Google Colab cloud-based environment, utilizing GPU acceleration when available to improve computational efficiency. The Torchvision library was used to access pretrained convolutional neural network architectures and perform image preprocessing operations, while Scikit-learn provided utilities for dataset partitioning and evaluation metrics [15]. Additional libraries included NumPy for numerical computation, Pillow (PIL) for image processing, Matplotlib for visualization of learning curves, and Seaborn for confusion matrix visualization.

A. Dataset Configuration

The experimental dataset consisted of facial image frames extracted from the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) [11]. Since the objective of this study was binary facial emotion classification, the original multi-class emotion labels were transformed into two categories. Images representing the “happy” emotion were assigned to the positive class, whereas all remaining emotions were grouped into the “not happy” class.

To reduce classification bias, the majority class was randomly downsampled to match the number of samples in the minority class, producing a balanced dataset of 2,332 facial images. The dataset was partitioned using an 80/20 stratified split, resulting in 1,865 images for model development and 467 images reserved for independent testing. The holdout test set remained completely unseen during model training and selection.

B. Image Preprocessing

Image preprocessing and data augmentation were implemented using Torchvision transforms [12]. All images were

resized to the input dimensions required by the pretrained convolutional neural networks and normalized using the ImageNet mean and standard deviation to maintain compatibility with pretrained weights [7].

To improve model robustness and reduce overfitting, data augmentation was applied exclusively to the training data. Augmentation operations included random horizontal flipping, random image rotation, and color jittering. Validation and test images received only deterministic resizing and normalization to ensure fair performance evaluation.

C. Network Architectures

Two transfer learning models were evaluated in this work: ResNet18 [9] and MobileNetV2 [10]. Both architectures were initialized using ImageNet pretrained weights [7] and adapted for binary classification by replacing the original classification layer with a two-node fully connected output layer. Fine-tuning was then performed on the target dataset following standard transfer learning practices [6].

ResNet18 represents a residual learning architecture that utilizes skip connections to facilitate optimization of deeper neural networks [9]. In contrast, MobileNetV2 employs inverted residual blocks and linear bottlenecks to achieve high classification accuracy while substantially reducing computational complexity, making it suitable for mobile and edge computing applications [10], [2].

D. Training Configuration

Model optimization was performed using the Adam optimization algorithm [14] with an initial learning rate of 1×10^{-4} , a batch size of 32, and the cross-entropy loss function for binary classification. Each model was trained for a maximum of 15 epochs.

A ReduceLROnPlateau learning rate scheduler monitored the validation loss throughout training and automatically reduced the learning rate whenever convergence plateaued. To prevent overfitting, early stopping with a patience of three consecutive epochs was employed. The model achieving the lowest validation loss within each fold was retained for subsequent evaluation.

E. Cross-Validation and Independent Testing

Model performance was assessed using stratified five-fold cross-validation, which preserves class proportions within every fold while providing reliable estimates of model generalization performance [13]. During each iteration, four folds were used for model training and one fold was reserved for validation.

After completion of all five folds, the model achieving the lowest validation loss across the entire cross-validation procedure was selected as the final model. This model was subsequently evaluated on the independent holdout test set consisting of 467 previously unseen images. The combination of stratified cross-validation and independent testing provides a robust evaluation methodology that minimizes bias and produces reliable estimates of real-world performance.

F. Performance Evaluation

Model performance was evaluated using multiple complementary classification metrics, including accuracy, precision, recall, and F1-score. In addition, confusion matrices and detailed classification reports were generated to analyze prediction behavior and classification errors. All performance metrics were computed using the Scikit-learn machine learning library [15].

Final performance was reported as the average validation metrics obtained across the five cross-validation folds together with the results achieved on the independent holdout test set. This evaluation protocol provides a comprehensive assessment of model accuracy, stability, and generalization capability while enabling a direct comparison between ResNet18 and MobileNetV2.

V. RESULTS AND DISCUSSION

The proposed facial emotion recognition framework was evaluated using the RAVDESS image dataset under a binary classification setting (Happy vs. Not Happy) [11]. Two transfer learning models, ResNet18 and MobileNetV2, pretrained on ImageNet, were trained using five-fold stratified cross-validation and subsequently evaluated on an independent holdout test set [7], [6], [13]. Model performance was assessed using accuracy, precision, recall, F1-score, and confusion matrices, which are standard evaluation metrics for classification tasks [1].

A. Cross-Validation Performance

Tables I and II summarize the validation performance obtained by each model across the five folds.

TABLE I
RESNET18 VALIDATION PERFORMANCE

Fold	Accuracy (%)	Precision	Recall	F1-score
1	93.30	0.9765	0.8877	0.9300
2	96.51	0.9485	0.9840	0.9659
3	96.51	0.9439	0.9893	0.9661
4	98.12	0.9637	1.0000	0.9815
5	90.35	0.9934	0.8118	0.8935
Average	94.96	0.9652	0.9346	0.9474

The ResNet18 model demonstrated strong and consistent performance across folds. Accuracy ranged from 90.35% to 98.12%, with Fold 4 achieving perfect recall, indicating all positive samples were correctly classified. However, Fold 5 showed reduced recall, suggesting higher false negatives. Overall, the model generalised well across splits due to residual learning capabilities [9].

MobileNetV2 achieved stable performance across all folds. Although its precision was slightly lower than ResNet18, it achieved higher recall on average, indicating improved sensitivity to positive class detection. This reflects the efficiency of depthwise separable convolutions in MobileNetV2 for feature extraction [10].

Overall, both architectures demonstrated strong generalisation ability, indicating that transfer learning effectively reduces overfitting in limited data scenarios [6].

TABLE II
MOBILENETV2 VALIDATION PERFORMANCE

Fold	Accuracy (%)	Precision	Recall	F1-score
1	94.64	0.9196	0.9786	0.9482
2	96.25	0.9391	0.9893	0.9635
3	94.64	0.9034	1.0000	0.9492
4	96.78	0.9394	1.0000	0.9688
5	93.83	0.9711	0.9032	0.9359
Average	95.23	0.9345	0.9742	0.9531

B. Holdout Test Performance

After cross-validation, the best-performing models were evaluated on an independent test set of 467 samples.

TABLE III
FINAL TEST PERFORMANCE

Model	Accuracy (%)	Precision	Recall	F1-score	Loss
ResNet18	96.15	0.9317	0.9957	0.9627	0.1412
MobileNetV2	96.36	0.9463	0.9828	0.9642	0.1007

Both models achieved strong performance on unseen data, exceeding 96% accuracy. ResNet18 produced the highest recall, making it highly sensitive to positive class detection. However, MobileNetV2 achieved better overall balance across all metrics, including the lowest loss value, indicating more stable predictions.

C. Comparative Analysis

TABLE IV
OVERALL MODEL COMPARISON

Model	Val Acc (%)	Test Acc (%)	Precision	Recall	F1-score
ResNet18	94.96	96.15	0.9317	0.9957	0.9627
MobileNetV2	95.23	96.36	0.9463	0.9828	0.9642

Both architectures successfully adapted to the facial emotion recognition task using transfer learning from ImageNet-pretrained weights [7]. MobileNetV2 consistently achieved slightly better overall performance, while ResNet18 excelled in recall. These results align with prior work showing that lightweight architectures can achieve competitive accuracy with improved efficiency [2].

Overall, MobileNetV2 is more suitable for balanced performance and deployment scenarios, while ResNet18 is preferable in applications where recall is critical.

The results demonstrate that transfer learning with convolutional neural networks provides an effective solution for binary facial emotion recognition using the RAVDESS dataset, achieving approximately 96% accuracy on unseen data [11].

VI. CONCLUSION

This work presented a comparative study of two convolutional neural network architectures, ResNet18 and MobileNetV2, for facial emotion recognition using transfer learning from ImageNet-pretrained weights [7]. The models were trained and evaluated using a 5-fold cross-validation strategy

and a final held-out test set, ensuring robust and reliable performance estimation [13].

Experimental results demonstrate that both architectures achieve strong and competitive performance, with MobileNetV2 slightly outperforming ResNet18 in terms of overall test accuracy and F1-score. Specifically, MobileNetV2 achieved a final test accuracy of 96.36%, while ResNet18 achieved 96.15%. This indicates that lightweight architectures can effectively match deeper networks when properly optimized, while offering reduced computational cost [10], [2].

The success of both models highlights the effectiveness of transfer learning for visual emotion recognition tasks, particularly in scenarios with limited labeled data [6]. Residual learning in ResNet18 enables strong feature representation [9], while depthwise separable convolutions in MobileNetV2 provide an efficient alternative with comparable accuracy.

Optimization using Adam [14], combined with standardized deep learning frameworks such as PyTorch [12] and Scikit-learn [15], contributed to stable training and reproducible evaluation.

In conclusion, MobileNetV2 is better suited for real-time and resource-constrained environments, while ResNet18 remains a strong baseline for high-capacity feature learning. Future work could explore transformer-based architectures and multimodal fusion approaches to further enhance robustness in real-world emotion recognition systems [5].

REFERENCES

- [1] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.
- [2] V. Sze, Y.-H. Chen, T.-J. Yang, and J. Emer, "Efficient processing of deep neural networks: A tutorial and survey," *Proceedings of the IEEE*, vol. 105, no. 12, pp. 2295–2329, 2017.
- [3] Y. Liu, J. Sun *et al.*, "Efficient deep convolutional neural networks for edge ai: A 2024 survey," *Pattern Recognition*, 2024.
- [4] W. Zhang, H. Chen *et al.*, "Recent advances in transfer learning for deep neural networks: A comprehensive survey," *IEEE Access*, 2024.
- [5] B.-K. Ko, "A brief review of facial emotion recognition based on visual information," *Sensors*, vol. 18, no. 2, 2018.
- [6] S. J. Pan and Q. Yang, "A survey on transfer learning," *IEEE Transactions on Knowledge and Data Engineering*, vol. 22, no. 10, pp. 1345–1359, 2010.
- [7] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "Imagenet: A large-scale hierarchical image database," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2009, pp. 248–255.
- [8] H. Wang, R. Li *et al.*, "Facial expression recognition in the era of deep learning: A 2024 survey," *Information Sciences*, 2024.
- [9] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 770–778.
- [10] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "Mobilenetv2: Inverted residuals and linear bottlenecks," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 4510–4520.
- [11] S. R. Livingstone and F. A. Russo, "The ryerson audio-visual database of emotional speech and song (ravdess): A dynamic, multimodal set of facial and vocal expressions in north american english," *PLoS ONE*, vol. 13, no. 5, 2018.
- [12] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury *et al.*, "Pytorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.
- [13] R. Kohavi, "A study of cross-validation and bootstrap for accuracy estimation and model selection," *Proceedings of the International Joint Conference on Artificial Intelligence*, pp. 1137–1143, 1995.
- [14] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *International Conference on Learning Representations*, 2015.
- [15] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg *et al.*, "Scikit-learn: Machine learning in python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.